EC991 Topics in Macroeconomics

Week 5 Tutorial · Solving Hopenhayn

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Baseline Hopenhayn model

We now take the model to the computer. The ingredients, in the notation of the code, with the wage as numeraire, $w = 1$:

- Profit of a firm with productivity z at price p, after choosing labour $n(z;\, p)$:

$$\pi(z;\, p) \;=\; (1-\eta)\,(p\,z)^{\frac{1}{1-\eta}}\;\eta^{\frac{\eta}{1-\eta}} \;-\; c_f$$

- Incumbents decide whether to operate next period; entrants pay c_e and draw z from G:

$$V(z) \;=\; \pi(z;\, p) + \beta\,\max\Big\{\int V(z')\,dF(z' \mid z),\; 0\Big\}, \qquad V^e \;=\; -c_e + \beta\int V(z)\,dG(z)$$

- We must solve for the price p and the mass of entrants M such that **free entry** holds and the **goods market clears**:

$$V^e(p) \;=\; 0, \qquad D(p) \;=\; \int y(z;\, p)\,d\mu(z)$$

where μ is the stationary measure of operating firms, which scales with M

Algorithm in a nutshell

Discretise z on n_z grid points, with transition matrix F and entrants' probabilities g. The solution has two blocks: the **price** from free entry, then the **mass of entrants** from the goods market

Step 1. Guess a price p^0. Compute $\pi(z_i;\, p^0)$, $n(z_i;\, p^0)$ and $y(z_i;\, p^0)$ at every grid point

Step 2. Solve the incumbent's Bellman equation by value function iteration, and record the exit rule χ:

$$V(z_i) \;=\; \pi(z_i;\, p^0) + \beta\, \max\Big\{ \sum_j F_{ij}\, V(z_j),\; 0 \Big\}, \qquad \chi(z_i) \;=\; \mathbf{1}\Big\{ \sum_j F_{ij}\, V(z_j) < 0 \Big\}$$

Step 3. Compute the value of entry:

$$V^e(p^0) \;=\; -c_e + \beta \sum_i g_i\, V(z_i;\, p^0)$$

Step 4. If $V^e(p^0) \neq 0$, update the price (lower it if $V^e > 0$, raise it if $V^e < 0$) and return to Step 1

Step 5. At the equilibrium price, compute the stationary measure with $M = 1$, where $\hat{F}_{ij} = (1 - \chi(z_i))\, F_{ij}$:

$$\bar{\mu} \;=\; (I - \hat{F}')^{-1} g$$

Step 6. Clear the goods market for M, then scale up the measure:

$$M \;=\; \frac{D(p)}{\sum_i y(z_i;\, p)\, \bar{\mu}_i}, \qquad \mu \;=\; M\bar{\mu}, \qquad B \;=\; M \sum_i \bar{\mu}_i$$

Which week 4 tools carry over?

- EGM needs a continuous choice whose first-order condition links today and tomorrow. Here the intertemporal decision is "stay or go": there is nothing to invert
- EGM returns when firms make a continuous dynamic choice, such as investment with capital (see the conclusion of the lecture)

Week 4 tool	**Here?**	**Why**
Endogenous grid method	**no**	no Euler equation to invert: labour is chosen statically, in closed form, and the only dynamic choice is discrete
Howard's improvement	yes	hold the exit rule fixed and update V cheaply
Policy iteration	yes	with χ fixed, V solves a **linear** system exactly
Bisection on a price	yes	$V^e(p)$ is monotone, as $\Phi(r)$ was
A distribution on a grid	yes	$\mu = (I - \hat{F}')^{-1} Mg$, or iterate the law of motion
A lottery between grid points	yes	split the grid cell where $\tilde{z}$ falls between exit and stay

Howard acceleration: hold the exit rule fixed

- Start with a value guess $V^0(z_i) = \pi(z_i;\, p)$
- Use expected continuation values to choose an exit rule:

$$\chi(z_i) \;=\; \begin{cases} 1 & \text{if } \sum_j F_{ij}\, V^0(z_j) < 0 \\ 0 & \text{otherwise} \end{cases}$$

- **Hold this rule fixed for n_H value updates:**

$$V^{k+1}(z_i) \;=\; \pi(z_i;\, p) + \beta\,[1 - \chi(z_i)] \sum_j F_{ij}\, V^k(z_j)$$

- At exit states, value is today's profit. At stay states, add discounted continuation value
- After n_H updates, **reconsider the exit rule** using the latest values. Repeat until convergence
- **What is saved?** We avoid choosing between stay and exit at every value update

Policy iteration: solve the fixed-rule value directly

- Start with an exit rule χ, for example the rule obtained from $V^0 = \pi$
- **Evaluate it fully:** solve the simultaneous equations for the value of following that rule:

$$\left[I - \beta \operatorname{diag}(1 - \chi) F\right] V \;=\; \pi$$

- **Improve it:** compute FV; exit where continuation value is negative and stay otherwise
- If the rule changes, solve again. **Stop when the evaluated rule remains optimal**
- **Does this save computing time here?**
  - All methods give the same equilibrium price at each grid size
  - **Here, VFI is fastest:** the stay/exit comparison is cheap; Howard still computes FV, while policy iteration requires a linear solve

Method	$n_z = 101$	$n_z = 1001$
VFI	0.004 s	0.492 s
Howard, $n_H = 20$	0.005 s	0.604 s
Policy iteration	0.029 s	5.204 s

One thread on a 2.2 GHz server, after a warm-up. Your times will differ

An example

Example: parameters

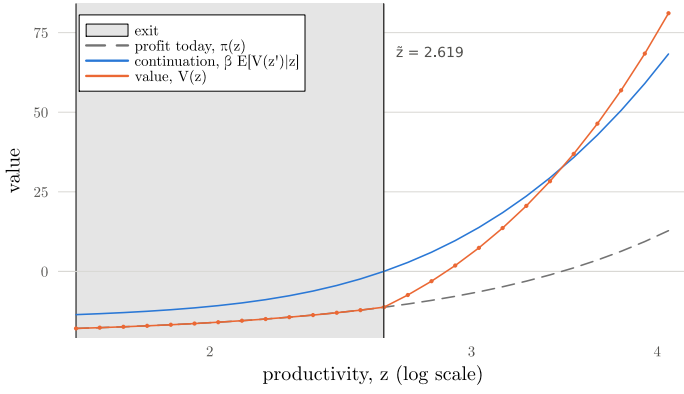
- Log productivity follows an AR(1) around a mean of 1:

$$\log z' - 1 \;=\; \rho\,(\log z - 1) + \sigma\,\varepsilon', \qquad \varepsilon' \sim N(0,1)$$

- Tauchen with $n_z = 101$ points, ± 4 standard deviations
- Entrants draw from the **stationary distribution** of the incumbents' chain, $g = f$

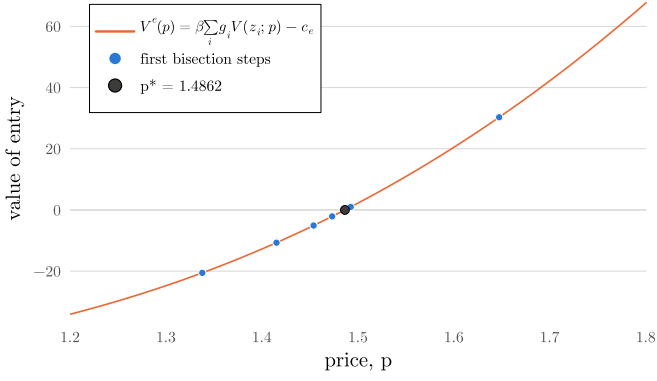
Parameter	Value	Role
η	$2/3$	curvature of the production function
β	0.8	discount factor
c_e	40	entry cost, in workers
c_f	20	fixed cost per period, in workers
$\bar{D}$	100	demand, $D(p) = \bar{D}/p$
ρ, σ	0.9, 0.2	persistence and volatility of productivity

Example: the incumbent's decision



- Below $\tilde{z}$ the continuation value is negative: the firm produces this period and leaves, so $V = \pi$
- Just above $\tilde{z}$ profit is still negative, but the firm stays: z is persistent but mean-reverting, and staying keeps the option of better draws

Example: the free entry price



- $V^e(p)$ is increasing: a higher price raises profit at every z. Bisection closes in on the one price where entry just breaks even
- The price does not depend on demand $\bar{D}$: free entry alone pins it down. Demand only scales the number of firms

Example: the stationary distribution



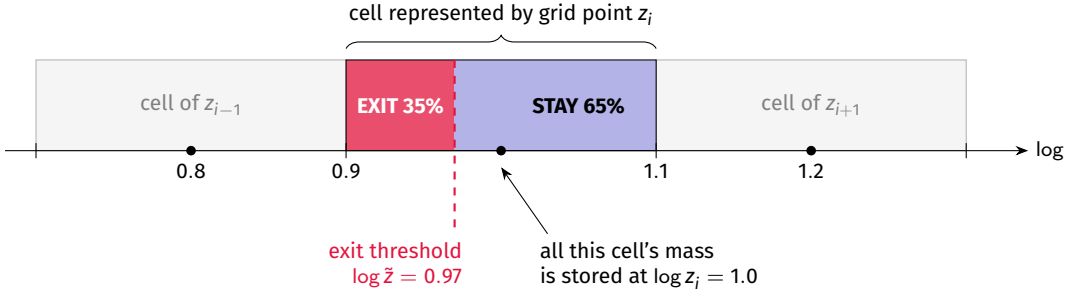
- Most firms are small, but most workers are in large firms: firms with 100 or more workers are 29% of firms and employ 73% of workers
- The distribution is cut off on the left: firms below the exit threshold leave after one period

Example: the equilibrium

- The **entry rate equals the exit rate**: in a stationary equilibrium, entrants replace exiting firms one for one
- Production labour is exactly $\eta \bar{D} = 66.7$: firms spend the share η of revenue $pY = \bar{D}$ on production workers

Equilibrium		**Size classes (workers)**		
			Firms	Employment
Price p	1.4862	<10	5.7%	0.3%
Mass of entrants M	0.0860	10–19	12.8%	1.9%
Number of firms	0.6413	20–49	29.5%	9.6%
Entry rate = exit rate	0.134	50–99	22.9%	15.3%
Exit threshold $\tilde{z}$	2.619	100–499	26.5%	49.5%
Average size (workers)	104.0	500+	2.7%	23.3%
Mean age (years)	11.5			
Output $Y = \bar{D}/p$	67.29			
Labour: production, fixed, entry	66.7, 12.8, 3.4			

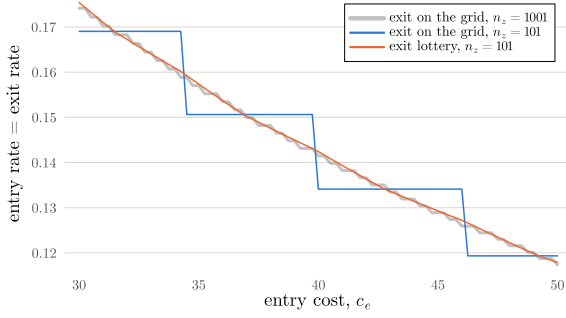
A grid point represents a productivity cell



- **Binary grid rule:** the representative point is above the threshold, so the whole cell stays
- **Continuous-model interpretation:** firms below the threshold should exit
- **Lottery correction:** assign 35% of the cell's mass to exit and 65% to survival

$$\text{Exit share} = \frac{0.97 - 0.9}{1.1 - 0.9} = 0.35$$

Discrete choices on a grid: exit rate as the entry cost rises



- **A sudden jump in a computed outcome may be a numerical artefact, not an economic prediction**
- The graph compares the binary rule on the coarse grid, the lottery on the coarse grid, and a fine grid. At the baseline, the lottery cuts the gap to the fine-grid exit rate from 0.8 to 0.1 percentage points
- On a given grid, the price is the same with and without the lottery: the lottery is applied only when computing the distribution, after the value function and the price are solved

Discrete choices on a grid: why the exit rate jumps

- **What should happen economically?**
  - As the entry cost c_e rises, the equilibrium price rises: incumbents become more profitable, and the exit threshold $\tilde{z}$ falls
  - Firms that used to exit now survive, so the exit rate should fall gradually as c_e rises
- **What happens with a binary decision at each grid point?** Take a grid point at $z_i = 3$:
  - Threshold at $\tilde{z} = 3.02$: every firm represented by point 3 exits
  - Threshold falls to 3.01: nothing changes, all those firms still exit
  - Threshold falls to 2.99: the whole group suddenly survives
  - The threshold moved only slightly, but **a whole grid point's mass switched from exit to survival**. Between such crossings the exit rule does not change, so the computed exit rate is a **staircase**
- **What does the lottery change?**
  - With the **exit lottery**, point 3 represents a productivity cell. As the threshold falls through the cell, the share assigned to exit falls gradually, for example from 60% to 55% to 50%, instead of the whole group switching at once
  - The exit rate curve is smoother, and closer to the one computed on a much finer grid

From the equilibrium measure to a simulated firm panel

- The solved model gives firm masses μ_i, total mass $B = \sum_i \mu_i$, and entry flow M. **These need not be integers**
- Choose a simulation size, say $N_{\text{sim}} = 10{,}000$. Preserve the equilibrium proportions:

$$\Pr(\text{operating firm is at } z_i) \;=\; \frac{\mu_i}{B}, \qquad \text{entrants per year} \;\approx\; N_{\text{sim}}\,\frac{M}{B}$$

- **Initialise:** start with an empty industry and introduce this number of entrants each year. Give each a new ID, age zero, and productivity drawn from g
- **Each year:** firms employ $n(z_i;\, p)$ workers and produce $y(z_i;\, p)$. Record their ID, age, productivity, employment and output
- **Exit and transition:** apply χ_i after production. Survivors keep their IDs, draw next productivity from row i of F, and age by one year
- **Burn in, then record:** allow the population to approach stationarity before collecting the panel. Firm counts fluctuate around N_{sim}; entry does not mechanically replace each realised exit

Backup: refining the grid

- The price and the interpolated threshold settle quickly: they depend on V, which is smooth in z
- The exit rate on the grid wanders until n_z is large; with the lottery it is within 0.001 of its limit from 101 points

		Entry rate = exit rate		
n_z	Price p	Exit on the grid	Exit lottery	Threshold $\tilde{z}$
51	1.48150	0.1417	0.1410	2.6148
101	1.48617	0.1341	0.1424	2.6188
201	1.48723	0.1382	0.1419	2.6172
501	1.48753	0.1407	0.1417	2.6168
1001	1.48758	0.1415	0.1417	2.6168
2001	1.48759	0.1419	0.1417	2.6168